BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 129 · Programming: Subprograms and Parameters

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How can a loop total an array's values?

Q2. Why is a thermostat controller an embedded system?
A. It has a specific task
B. It runs any desktop app
C. It always needs Wi-Fi
D. It always has a screen

Q3. “DNS assigns the same MAC address to all computers visiting a website.” Correct this. Explain why.

Q4. RAM is full. Apps keep moving data between RAM and an SSD. Correct both claims. Explain why. (a) Virtual memory needs no secondary storage. (b) Virtual memory is faster than RAM because SSDs hold more.

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I Do · Read new knowledge

Key vocabulary

Procedure — Runs a named group of instructions, often for a useful action.

Function — Returns a value that can be used in another expression.

Parameter — Allows a subprogram to receive values from its caller.

Modularity — Separating repeated tasks into subprograms reduces duplicated code.

New learning

Procedure: Runs a named group of instructions, often for a useful action.

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Worked example

Trace cube(3) for a function returning n*n*n.

1. Caller passes argument 3.

2. Parameter n receives 3.

3. Compute 3×3×3=27.

4. Return 27 to the caller.

We Do · Practise with guidance

Guided 1

A procedure printGreeting(name) displays 'Hello '+name for Mia.

Worked method: Output 'Hello Mia'.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the statement that returns the new two-item array.

```
function pickEnds(values)
    array pair[2]
    pair[0] = values[0]
    pair[1] = values[2]
    ...
endfunction
array sizes = [2,4,8]
result = pickEnds(sizes)
print(result[1])
```

write

Question 1

Explain one difference between a function and a procedure. [2 marks]

Question 2

State the role of parameters. [2 marks]

Question 3

Give one benefit of modular programming. [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Functions can return values to their callers.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Set a total to 0. Loop through the array, adding each value once.

Q2. A — It has a specific task

Q3. Distributed DNS servers help look up the IP address associated with a domain name.

Q4. (a) Virtual memory uses secondary storage when RAM is too full. (b) Moving data between RAM and slower secondary storage takes time. More storage space does not mean faster access.

I Do / We Do pages · Guided practice

Q1. Output 'Hello Mia'.

You Do pages · Code First

```
function pickEnds(values)
    array pair[2]
    pair[0] = values[0]
    pair[1] = values[2]
    return pair
endfunction
array sizes = [2,4,8]
result = pickEnds(sizes)
print(result[1])
```

Expected result: 8

You Do pages · Independent questions

Q1. A function returns a value; a procedure performs statements without necessarily returning a value.

Q2. They pass input values into a subprogram.

Q3. Reduces duplication, simplifies testing or maintenance.

Exit Ticket · Model answer

True

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: